# CHAPTER - I

# INTRODUCTION

1.1 The present Report on Annual Survey of Industries (ASI) for the year 2010-11 is the thirty-eighth in the series of such reports brought out by this Directorate. It incorporates data in respect of industrial statistics collected through the Annual Survey of Industries in the organized factory sector in this State by the National Sample Survey Organization (NSSO), Government of India, under the statutory provisions of the collection of Statistics Act, 1953 and the rules framed there under in 1959.

1.2 The Director General, Field Operations Division (FOD), N.S.S.O., Government of India is declared as the Statistics Authority in terms of the aforesaid Act, while the Director of Planning, Statistics and Evaluation (DPSE), Government of Goa, has been appointed as the Officer on Special Duty for Goa State by the Government of India to ensure proper supervision and control over the coverage of factories located in this State.

1.3 **Scope and coverage:** The A.S.I. covers the factories registered under The Factories Act, 1948 on sample basis from the year 1999-2000 (accounting year of A.S.I. 1998-99), shifting from the existing pattern of selection of factories of complete enumeration or census in the State. The coverage includes manufacturing factories as also those service industries and activities like cold storage, water supply, repair of motor vehicles and other consumer durables such as watch, etc. As against 520 factories covered under Annual Survey of Industries during 2009-10, the numbers of factories covered during 2010-11 were 583.

1.4 **Classification of Industries**: The NIC-1970 was followed to classify economic activities of the factories from ASI 1973-74 to ASI 1988-89. NIC-1987 had then been introduced and followed till ASI 1997-98. NIC-1998 was then followed from ASI 1998-99 to ASI 2003-04. From ASI 2004-05, the new series of classification i.e. NIC-2004 has been introduced and the same has been used till ASI 2007-08. For ASI 2008-09 onwards, NIC-2008 has been introduced and used. It classifies all the factories in the ASI frame in their appropriate industry groups on the basis of the principal product manufactured. This way a unit gets classified in one and only one industry group even though it might be manufacturing products belonging to different industries. The estimates for different aggregates presented in this report at two or three or four digit level of industry correspond to the NIC-2008 classification. Details of this classification have been shown in Annexure- 8.

## CHAPTER - II

# CONCEPTS AND DEFINITIONS

Important concepts and definitions used in ASI are explained below.

**2.1** A number of accounting, economic and other terms related to the factories are used in conduct of Annual Survey of Industries. For proper assessment and interpretation of ASI data, those are given below.

**2.2 ACCOUNTING YEAR:** For the purpose of ASI, the accounting year is the period on which the factory closes its books of account. With the enactment of Income Tax Act, by and large, the accounting year of all factories is from April to March.

**2.3 REFERENCE PERIOD:** It is corresponds to the financial year. For example, for ASI 2010-2011 the reference period is the financial year commencing from 1st April 2010 and ending on 31st March 2011 or the accounting year of the factory ending on any date between 1.4.10 to 31.3.11.

**2.4 SURVEY PERIOD:** Survey period is a period during which work of any Annual Survey of Industries is undertaken. Since ASI 1978-79 the survey period has been fixed from 1st July to 30$^{th}$ June of the next year. From ASI 1998-99 the survey period has been changed from 12 months to 4 months i.e. from Nov'99 to Feb'2000. The survey period for ASI 99-00 has been again changed from October to March. The survey period for ASI 2010-2011 is from September, 2011 to April,2012.

**2.5 FACTORY (AS PER THE FACTORY ACT 1948):** Factory is one, which is registered under Sections 2m(i) and 2m(ii) of the Factories Act, 1948. The Sections 2m(i) and 2m(ii) refer to any premises including the precinct thereof (i) wherein ten or more workers are working, or were working on any day of the preceding twelve months, and in any part of which a manufacturing process is being carried on with the aid of power, or is ordinarily so carried on, or (ii) wherein twenty or more workers are working, or were working on any day of the preceding twelve months and in any part of which a manufacturing process is being carried on without the aid of power, or is ordinarily so carried on. Closed factories with fixed assets on site are also considered as registered factories till they are de-registered and removed from the live-register maintained by the Chief Inspector of Factories (CIF) in the State.

**2.6 MANUFACTURING PROCESS:** This is as per Section 2(k) of the Factories Act, 1948.

**2.7 GROSS VALUE OF PLANT AND MACHINERY:** Gross value of plant and machinery represents the total original (un-depreciated) value of installed plant and machinery as at the end of the accounting year. It includes the book value of own constructed plant and machinery, if installed, and the approximate value of rented-in plant and machinery as at the time of renting in but excludes the value of rented-out plant and machinery. Total value of all the plant and machinery acquired on hire-purchase basis is

also included. Thus it represents the gross value of plant and machinery engaged in production process.

**2.8 FIXED CAPITAL:** Fixed Capital represents the depreciated value of fixed assets owned by the factory as on the closing day of the accounting year. Fixed assets are those, which have normal productive life of more than one year. Fixed capital covers all type of assets, new or used or own constructed, deployed for productions, transportation, living or recreational facilities, hospitals, schools, etc. for factory personnel. It would include land, building, plant and machinery, transport equipment etc. It includes the fixed assets of the head office allocable to the factory and also the full value of assets taken on hire-purchase basis (Whether fully paid or not) excluding interest element. It excludes intangible assets and assets solely used for post manufacturing activities such as, sale, storage, distribution, etc.

**2.9 DEPRECIATION:** Depreciation is consumption of fixed capital by the factory due to wear and tear and obsolescence during the accounting year and is taken as provided by the factory owner, or if not provided by the factory this is estimated on the basis of cost of installation and working life of the fixed assets.

**2.10 FINISHED GOODS:** Finished Goods are those, which are manufactured by the factory for sale. Finished goods should conform to a prescribed standard.

**2.11 PHYSICAL WORKING CAPITAL:** This is defined to include all physical inventories owned, held or controlled by the factory as on the closing day of the accounting year such as the materials, fuels and lubricants, stores, etc. that enter into products manufactured by the factory itself or supplied by the factory to others for processing. Physical working capital also includes the value of stock of materials, fuels and stores etc. purchased expressly for re-sale, semi-finished goods and goods-in-process on account of others and goods made by the factory which are ready for sale at the end of the accounting year. However, it does not include the stock of the materials, fuels, stores, etc. supplied by others to the factory for processing. Finished goods processed by others from raw materials supplied by the factory and held by them are included and finished goods processed by the factory from raw materials supplied by others, are excluded.

**2.12 WORKING CAPITAL:** Working Capital is the sum total of the physical working capital as already defined above and the cash deposits in hand and at bank, land, the net balance of amounts receivable over amounts payable at the end of the accounting year. Amounts receivable include value of credit items on revenue account, such as sums due to the factory for goods sold, amounts advanced in connection with normal factory work, bills of exchange payable to the factory, payments made in advance such as for fire insurance, telephone charges, rates and taxes, call deposits and security deposits having a normal life of less than one year, etc. It excludes unused overdraft facility, fixed deposits irrespective of duration, advances for acquisition of fixed assets, long-term loans including interest thereon and investment.

**2.13 INVESTED CAPITAL:** Invested capital is the total of fixed capital and physical working capital.

**2.14 PRODUCTIVE CAPITAL:** This is the total of fixed capital and working capital.

**2.15 OUTSTANDING LOANS:** Outstanding loans represent all loans, whether short-term or long-term, whether interest bearing or not, outstanding according to the books of the factory as on the closing day of accounting year.

**2.16 CONTRACT WORKER:** All persons who are not employed directly by an employer but through the third agency, i.e. contractor, are termed as contract workers. Those workers may be employed with or without the knowledge of the principal employer.

**2.17 EMPLOYEES:** Employees relate to all persons engaged by the factory whether for wages or not, in work connected directly or indirectly with the manufacturing process and include all administrative, technical and clerical staff as also labour in production of capital assets for factory's own use. This is inclusive of persons holding position of supervision or management or engaged in administrative office, store-keeping section and welfare section, watch and ward staff, sales department as also those engaged in the purchase of raw materials etc. and production of fixed assets for the factory. It also includes all working proprietors and their family members who are actively engaged in the work of the factory even without any pay and the unpaid members of the co-operative societies who work in or for the factory in any direct and productive capacity. Persons in the head office connected with the manufacturing activity of the factory are also included in this item.

**2.18 LABOUR TURNOVER:** Labour turnover measures the extent of change in the working force due to accession and separation during a given period. The term 'accession' was defined as the total number of workers added to employment during the period, whether new or reemployed or transferred from other establishments or units under the same management. Interdepartmental transfers within the same establishment are, however ignored. The term 'separation' implies termination of employment at the instance of worker or employers. It includes termination of services due to death or retirement. As in the case of accession, transfers to other establishments are included but transfers within the same establishment are ignored. Retrenchment as a result of rationalisation or modernisation or any other cause, is also treated as separation.

**2.19 WAGES:** Wages are defined to include all remuneration capable of being expressed in monetary terms and also payable/paid more or less regularly in each pay period to workers (defined above) as compensation for work done during the accounting year. It includes:

(i) Direct wages and salary (i.e. basic wages/salaries, payment of overtime, dearness, compensatory, house rent and other allowances);

(ii) Remuneration for period not worked (i.e. basic wages), salaries and allowances payable for leave period, paid holidays, lay-off payments and compensation for unemployment (if not paid from source other than employers);

(iii) Bonus and ex-gratia payment paid both at regular and less frequent intervals (i.e., incentive bonuses and good attendance bonuses, production bonuses, profit sharing bonuses, festival or year end bonuses etc.).

It excludes layoff payments and compensation for employment except where such payments are for this purpose, i.e., payments not made by the employer. It excludes employer's contribution to old age benefits and other social security charges, direct expenditure on maternity benefits and creches and other group benefit in kind and travelling and other expenditure incurred for business purposes and reimbursed by the employer. The wages are expressed in terms of gross value, i.e., before deductions for fines, damages, taxes, provident fund, employee's state insurance contribution etc. Benefits in kind (perquisites) of individual nature are only included.

**2.20 BONUS:** Profit sharing bonus, festival bonus, year-end bonus, and all other bonuses and ex-gratia payments paid at less frequent intervals are covered by this term.

**2.21 WORKMEN AND STAFF WELFARE EXPENSES:** These include expenditure incurred by the employer on the maternity benefits and crèches and other benefits such as supply of food, beverages, tobacco, clothing and group lodging at confessional rates and educational, cultural and recreational facilities and services and grants to trade unions and cooperative stores meant for employees. All group benefits are included.

**2.22 EMOLUMENTS:** These are defined in the same way as wages but paid to all employees plus imputed value of benefits in kind i.e. the net cost to the employers on those goods and services provided to employees free of charge or at markedly reduced cost which are clearly and primarily of benefit to the employees as consumers. It includes profit sharing, festival and other bonuses and ex-gratia payments paid at less frequent intervals (i.e. other than bonus paid more or less regularly for each period). Benefits in kind include supplies or services rendered such as housing, medical, education and recreation facilities. Personal insurance, income tax, house rent allowance, conveyance etc. for payment by the factory also is included in the emoluments.

**2.23 SUPPLEMENTS TO EMOLUMENTS:** These include: (i) employer's contribution to old age benefits, i.e., provident fund, pension, gratuity, etc.; (ii) employer's contribution towards other social security charges such as Employees' State Insurance, compensation for work injuries, occupational diseases, maternity benefits, retrenchment and lay-off benefits etc.; and (iii) group benefits like direct expenditure on maternity, creches, canteen facilities, educational, cultural and recreational facilities and grant to trade unions, co-operative stores etc. meant for employees.

**2.24 COMPENSATION OF EMPLOYEES:** Compensation of employees is the total of emoluments and supplement to emoluments.

**2.25 MANDAYS WORKED:** These are obtained by summing up the number of mandays worked by persons working in each shift over all the shifts on all days, i.e. both manufacturing and non-manufacturing days. This figure excludes persons who are paid but remain on leave, strike, etc.

**2.26 MANDAYS PAID FOR:** The number of mandays paid for is arrived at by summing up the number of employees paid for in each shift. This also includes mandays on weekly

schedule holidays if paid for and those absences with pay as also mandays lost through lay off/ strike for which compensation was payable.

**2.27 WORKING DAY:** Working day means the days on which a manufacturing process and/ or repair or maintenance work was carried on.

**2.28 NON-WORKING DAY:** Apart from manufacturing day and repair and maintenance days there may be some non-working days. Non working days are those days on which the workers give their attendance but due to non-availability of raw materials power etc. no effective work is done. As the workers are paid for these days such days are also taken into account for the purpose of labour statistics.

**2.29 BASIC MATERIALS:** Basic materials are the materials which are important and of key nature to the industry on which the manufacturing process is based, vig. metal for machine, leather for shoe. Such material is not lost through the process of production but only changes its forms.

**2.30 CONSUMABLE STORES:** All such materials which assist the manufacturing process and loose their identity without entering the products are called consumable stores, e.g., cotton waste.

**2.31 FUEL CONSUMED:** Fuel Consumed represent total purchase value of all items of fuels, lubricants, electricity, water (purchased to make steam) etc. consumed by the factory during the accounting year except those which directly enter into products as materials consumed. It excludes that part of fuels, which is produced and consumed by the factory in manufacture i.e., all intermediate products and also fuels consumed by employees as part of amenities. It includes quantities acquired and consumed from allied concerns, their book value being taken as their purchase value and also the quantities consumed in production of machinery or other capital items for factory's own use.

**2.32 MATERIALS CONSUMED:** Materials consumed represent the total delivered value of all items of raw materials, components, chemicals, packing materials and stores which actually entered into the production process of the factory during the accounting year. It also includes the cost of all the materials used in the production of fixed assets, including construction work for factory's own use. Components and accessories fitted as purchased with the finished product during the accounting year are also to be included. It excludes intermediate products. Intermediate products in the above context mean all those products which are produced by the factory and consumed for further manufacturing process.

**2.33 TOTAL INPUT:** This comprises gross value of fuel materials etc. consumed (as defined above) and also other inputs viz. (a) cost of non-industrial services received from others (b) cost of materials consumed for repair and maintenance of factory's fixed assets including cost of work done by others to the factory's fixed assets (c) cost of contract and commission work done by others on materials supplied by the factory (d) cost of office supplies and products reported for sale during last year & used for further manufacture during the accounting year.

**2.34 INTERMEDIATE PRODUCT:** Intermediate Product is a product which is obtained during a manufacturing process, which may or may not be saleable and is not the intended final product.

**2.35 NET VALUE OF SEMI-FINISHED GOODS:** It represents the excess/deficit of value of semi-finished goods and/or goods-in-process at the end of the accounting year over that at the beginning of year.

**2.36 PRODUCTS:** These are defined to include the ex-factory value (i.e. exclusive of taxes, duties etc. on sale and inclusive of subsidies etc., if any) of all products and by-products, excluding intermediate products, that have been completed during the accounting year for sale whether actually sold during the accounting year or entered into books. Also include fixed assets produced by the factory for its own use.

**2.37 GROSS OUTPUT:** Gross output is defined to include the ex-factory value, (i.e., exclusive of taxes, duties, etc. on sale and inclusive of subsidies etc., if any) of products and by-products manufactured during the accounting year, and the net value of the semi-finished goods, work-in process, (represents the excess/deficit of value of semi-finished goods or work-in-process at the end of the accounting year over that of the beginning of the year plus net balance of semi-finished fixed assets on factory's capital account) and also the receipts for industrial and non industrial services rendered to others, value of semi-finished goods of last year sold in the current year, sale value of goods sold in the same condition as purchased and value of electricity generated and sold. Value of gross output and total output has been used in the text interchangeable to mean the same thing.

**2.38 INDUSTRIAL SERVICES:** Any services taken or rendered from one to another unit resulting in increase in the value of material during the manufacturing process are industrial services.

**2.39 NON-INDUSTRIAL SERVICES:** All such services which do not have a direct bearing on the manufacturing process but are needed by any manufacturing unit are called non-industrial services, say, transport.

**2.40 NET VALUE ADDED:** This is the increment to the value of goods and services that is contributed by the factory and is obtained by deducting the value of total inputs and depreciation from gross value of output.

# CHAPTER – III

# PRINCIPAL CHARACTERISTICS BY MAJOR INDUSTRIAL ACTIVITY

3.1 **Principal Aggregates** : The distribution of principal characteristics by major economic activity in absolute and percentage terms by the manufacturing and others sector of industrial activity in Goa covered by the Annual Survey of Industries (ASI) 2010-11 compared with 2009-10 are given below :

**TABLE NO. 3.1**

**PRINCIPAL CHARACTERISTICS BY MAJOR ECONOMIC ACTIVITY**

| Principal Characteristics | Unit | Manufacturing | | Others | | All | |
|---|---|---|---|---|---|---|---|
| | | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| | | | | | | | |
| Factories | Number | 498 | 555 | 22 | 28 | 520 | 583 |
| | (%) | (96) | (95) | (4) | (5) | (100) | (100) |
| Fixed Capital | Rs. in crore | 6295 | 5863 | 269 | 358 | 6564 | 6221 |
| | (%) | (96) | (94) | (4) | (6) | (100) | (100) |
| Invested Capital | Rs. in crore | 12934 | 11294 | 278 | 430 | 13212 | 11724 |
| | (%) | (98) | (96) | (2) | (4) | (100) | (100) |
| Workers | Number | 37871 | 34613 | 1020 | 1463 | 38891 | 36076 |
| | (%) | (97) | (96) | (3) | (4) | (100) | (100) |
| Total persons engaged | Number | 50397 | 50905 | 1300 | 2247 | 51697 | 53152 |
| | (%) | (97) | (96) | (3) | (4) | (100) | (100) |
| Wages to workers | Rs. in crore | 485 | 542 | 8 | 20 | 493 | 562 |
| | (%) | (98) | (96) | (2) | (4) | (100) | (100) |
| Total Emoluments | Rs. in crore | 1068 | 1212 | 15 | 53 | 1083 | 1265 |
| | (%) | (99) | (96) | (1) | (4) | (100) | (100) |
| Input | Rs. in crore | 22884 | 25273 | 341 | 667 | 23225 | 25940 |
| | (%) | (99) | (97) | (1) | (3) | (100) | (100) |
| Gross Output | Rs. in crore | 28994 | 31666 | 441 | 828 | 29435 | 32494 |
| | (%) | (99) | (97) | (1) | (3) | (100) | (100) |
| Depreciation | Rs. in crore | 550 | 697 | 19 | 30 | 569 | 727 |
| | (%) | (97) | (96) | (3) | (4) | (100) | (100) |
| Net Value Added | Rs. in crore | 5560 | 5697 | 80 | 131 | 5640 | 5828 |
| | (%) | (99) | (98) | (1) | (2) | (100) | (100) |
| Net Income | Rs. in crore | 5129 | 5177 | 68 | 112 | 5197 | 5289 |
| | (%) | (99) | (98) | (1) | (2) | (100) | (100) |

*Figures in bracket indicate percentages*.

3.2 The number of factory units covered under the Annual Survey of Industries during 2009-10 and 2010-11 were 520 and 583 units respectively. It may be observed from Table 3.1, that 498 factory units were engaged in the manufacturing activities in the year 2009-10, which has increased to 555

during the year 2010-11. The units covered under “other “industrial activities have also increased from 22 in 2009-10 to 28 in 2010-11.

3.3 It may be observed that the share of Manufacturing Sector in the total fixed capital has reduced from 96% in 2009-10 to 94% in 2010-11, i.e. (Rs.6295 crores) in the year 2009-10 and (Rs.5863 crores) during the year 2010-11. The percentage of fixed capital accounted for by the factories covered under the others group of industries was 4% in 2009-10 and 6% in 2010-11. The number of workers employed in the manufacturing has decreased from 97% in 2009-10 to 96% in 2010-11 and were paid wages (Rs.485 crores) during 2009-10 and (Rs.542 crores) during 2010-11 i.e. 98% and 96% during 2009-10 and 2010-11 respectively. The Gross output produced under manufacturing sector was valued at Rs.28994 crores & Rs.31666 crores or 99% in 2009-10 and 97% in 2010-11. The contribution of Net Value Added by manufacturing was Rs.5560 crores and Rs.5697 crores i.e 99% and 98% during the year 2009-10 and 2010-11 respectively.

The percentage growth during 2010-11 over 2009-10 in respect of important characteristics is exhibited in table 3.2 below.

**TABLE NO. 3.2**

**PERCENTAGE GROWTH DURING 2010-11 OVER 2009-10 IN PRINCIPAL CHARACTERISTICS BY MAJOR ECONOMIC ACTIVITY**

| Principal Characteristics | Manufacturing | Others | All Industries |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| Factories | 11.4 | 27.3 | 12.1 |
| Fixed Capital | -6.9 | 33.1 | -5.2 |
| Invested Capital | -12.7 | 54.7 | -11.3 |
| Workers | -8.6 | 43.4 | -7.2 |
| Total persons engaged | 1.0 | 72.8 | 2.8 |
| Wages to workers | 11.8 | 150.0 | 14.0 |
| Total Emoluments | 13.5 | 253.3 | 16.8 |
| Input | 10.4 | 95.6 | 11.7 |
| Gross Output | 9.2 | 87.8 | 10.4 |
| Depreciation | 26.7 | 57.9 | 27.8 |
| Net Value Added | 2.5 | 63.8 | 3.3 |
| Net Income | 1.0 | 64.7 | 1.8 |

**3.4 Growth**: It is observed from table no. 3.2 that total number of factory units has increased by (12.1%) during 2010-11 over 2009-10. The growth in Net Value Added during 2010-11 over 2009-10 under All industries was 3.3%. Among other characteristics enjoying positive growth in manufacturing sector are Total persons engaged (1.0%), Wages to workers (11.8%), Total Emoluments (13.5%), Input (10.4%), Gross output (9.2%), Depreciation (26.7%), Net Value Added (2.5%) and Net Income (1.0%). Whereas Fixed capital, Invested capital and Workers have shown negative growth of (-6.9%), (-12.7%) and (-8.6%) respectively.

**TABLE NO. 3.3**

**STRUCTURAL RATIOS AND TECHNICAL COEFFICIENTS BY MAJOR ECONOMIC ACTIVITY**

| Item | Unit | Manufacturing | | Others | | All | |
|---|---|---|---|---|---|---|---|
| | | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| **Structural Ratios** | | | | | | | |
| Fixed capital per factory | Rs. in lakh | 1264.06 | 1056.33 | 1222.73 | 1280.18 | 1262.3 | 1067.08 |
| Gross output per factory | Rs. in lakh | 5822.09 | 5705.57 | 2004.55 | 2957.86 | 5660.58 | 5573.60 |
| Net value added per factory | Rs. in lakh | 1116.47 | 1026.38 | 363.34 | 468.25 | 1084.62 | 999.57 |
| Workers per factory | Number | 76 | 62 | 46 | 52 | 75 | 62 |
| Persons engaged per factory | Number | 101 | 92 | 59 | 80 | 99 | 91 |
| Fixed capital per person engaged | Rs. in lakh | 12.5 | 11.52 | 20.7 | 15.95 | 12.7 | 11.70 |
| Gross output per person engaged | Rs. in lakh | 57.5 | 62.21 | 33.9 | 36.86 | 56.9 | 61.13 |
| Net value added per person engaged | Rs. in lakh | 11 | 11.19 | 6.6 | 5.83 | 10.9 | 10.96 |
| Total emoluments per person engaged | Rs. in lakh | 2.12 | 2.38 | 1.2 | 2.37 | 2.1 | 2.38 |
| Wages per worker | Rs. in lakh | 1.3 | 1.57 | 0.8 | 1.35 | 1.3 | 1.56 |
| **Technical Coefficient** | | | | | | | |
| Fixed capital to Net value added | | 1.13 | 1.03 | 3.36 | 2.73 | 1.16 | 1.07 |
| Fixed capital to Gross output | | 0.22 | 0.19 | 0.61 | 0.43 | 0.22 | 0.19 |
| Net value added to Gross output | | 0.19 | 0.18 | 0.18 | 0.16 | 0.19 | 0.18 |
| Total emoluments to Net value added | | 0.19 | 0.21 | 0.19 | 0.41 | 0.19 | 0.22 |

3.5 **Some technical structural ratios/coefficients**: Above table reveals certain important structural ratios and coefficients which is the basic structure of the contribution to the economy.

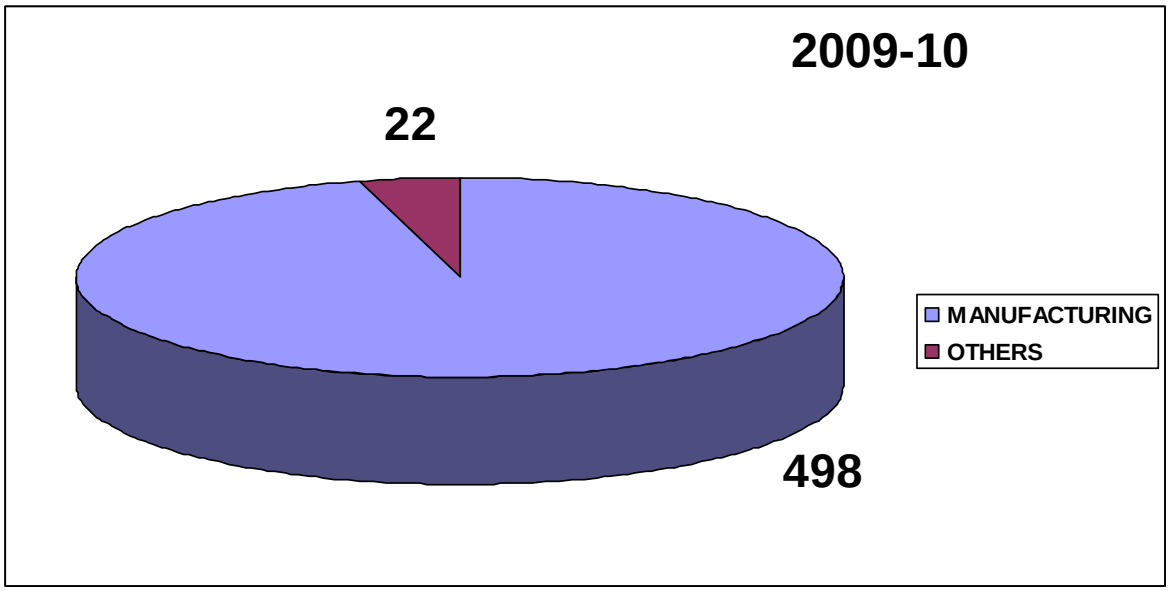
The fixed capital utilized by each factory unit in case of all the activities and that under manufacturing sector, which also forms the major sector, works out to the value of Rs.1262.3 lakh and Rs.1067.08 lakh and Rs.1264.06 lakh and Rs.1056.33 lakh respectively during 2009-10 & 2010-11. The ratio of Net value added per factory has declined from Rs.1116.47 lakhs in 2009-10 to Rs.1026.38 lakhs in 2010-11 under manufacturing sector whereas, it has increased from Rs.363.34 lakh in 2009-10 to Rs. 468.25 lakh in 2010-11 under “Others” category. Net value added per person engaged shows an increasing trend in manufacturing sector and decreasing trend under “Others”

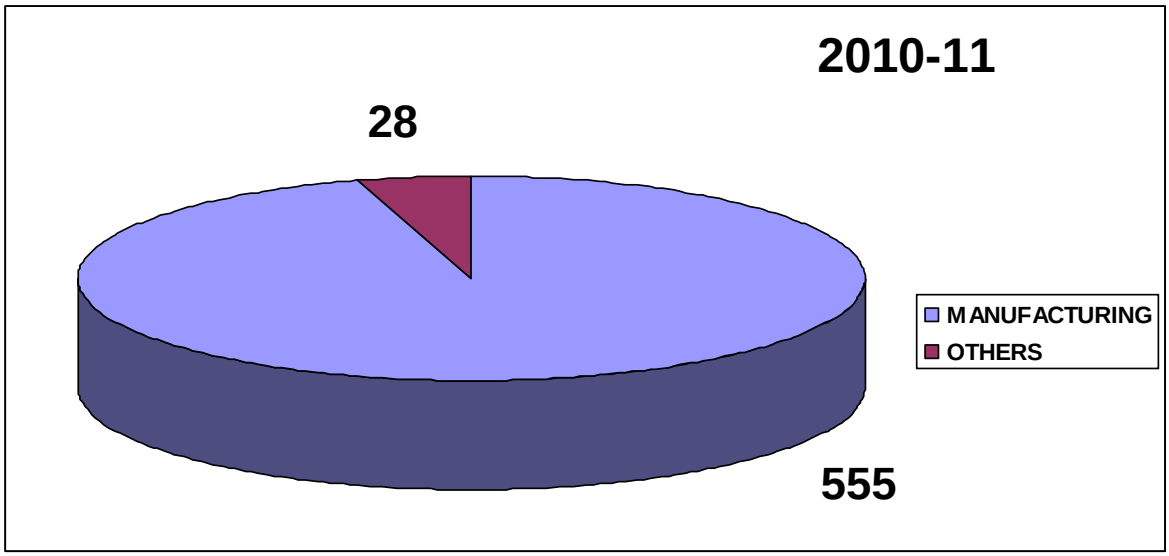
category of industries. It is important to note that the average wages per worker per annum has increased significantly from Rs.1.3 lakh in 2009-10 to Rs.1.57 lakhs in 2010-11. On the other hand, number of workers employed per factory during 2009-10 over 2010-11 has increased in factories covered under “others” group and decreased in the manufacturing sector. Gross output per person engaged shows an upward trend in manufacturing sector and “Others” sector.

As regards Technical Coefficients, the coefficient of fixed capital to net value added stood at 1.16 and 1.07 during 2009-10 and 2010-11 respectively under the sector “All activities”. The coefficient of Fixed Capital to Gross Output has shown a downward trend from 0.22 in 2009-10 to 0.19 in 2010-11. Also the coefficient of net value added to gross output has decreased from 0.19 for the year 2009-10 to 0.18 in the year 2010-11. Total emoluments paid in terms of Net value added generated has increased in Manufacturing sector and “Others” category from 0.19 and 0.19 in 2009-10 to 0.21 and 0.41 in 2010-11 respectively.

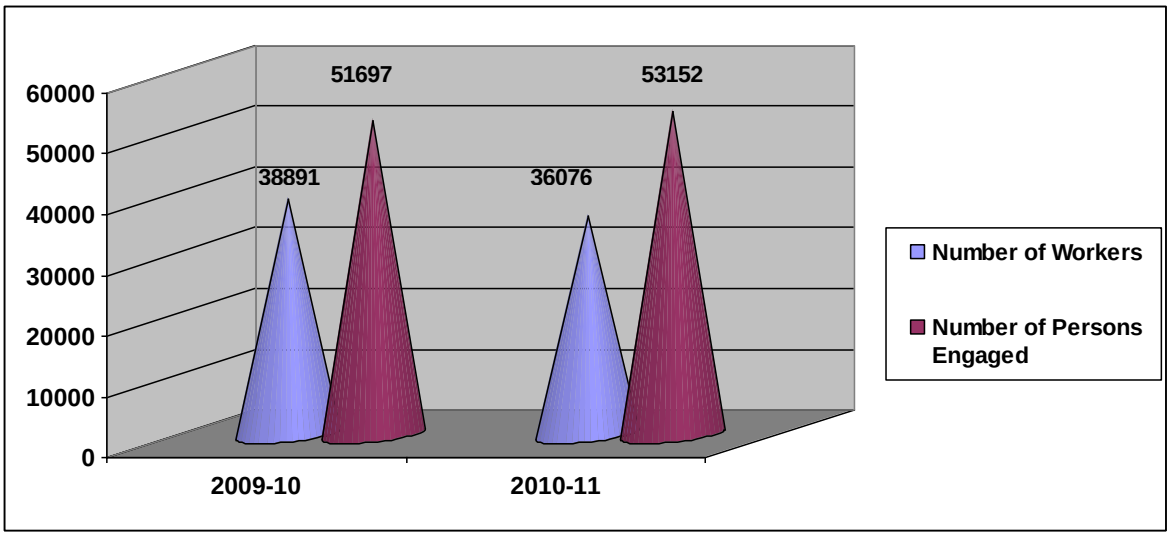
**PRINCIPAL CHARACTERISTICS BY MAJOR ECONOMIC ACTIVITY**

## (i) No. of Factories

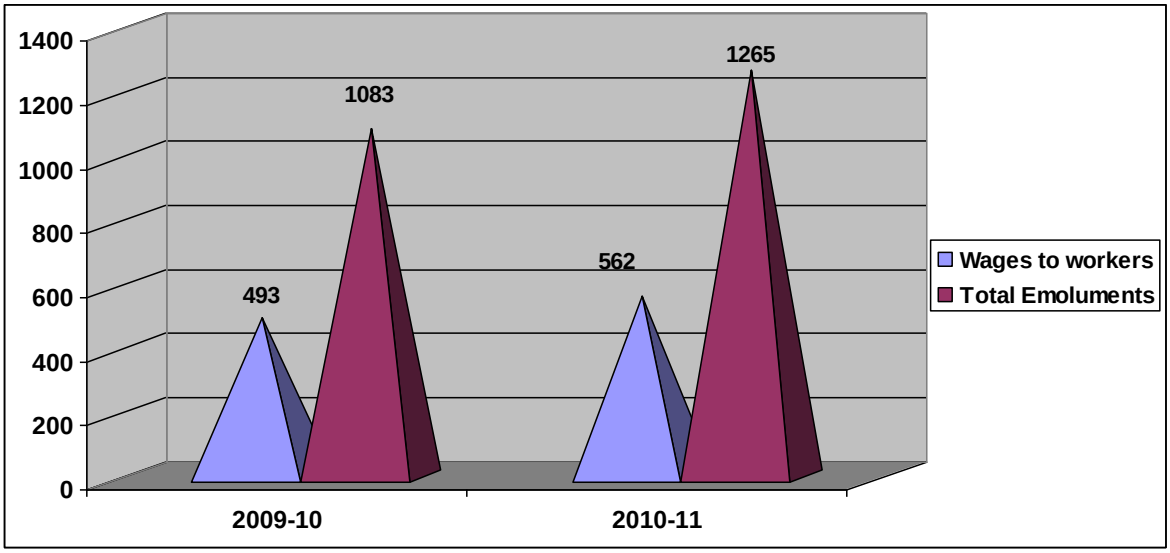


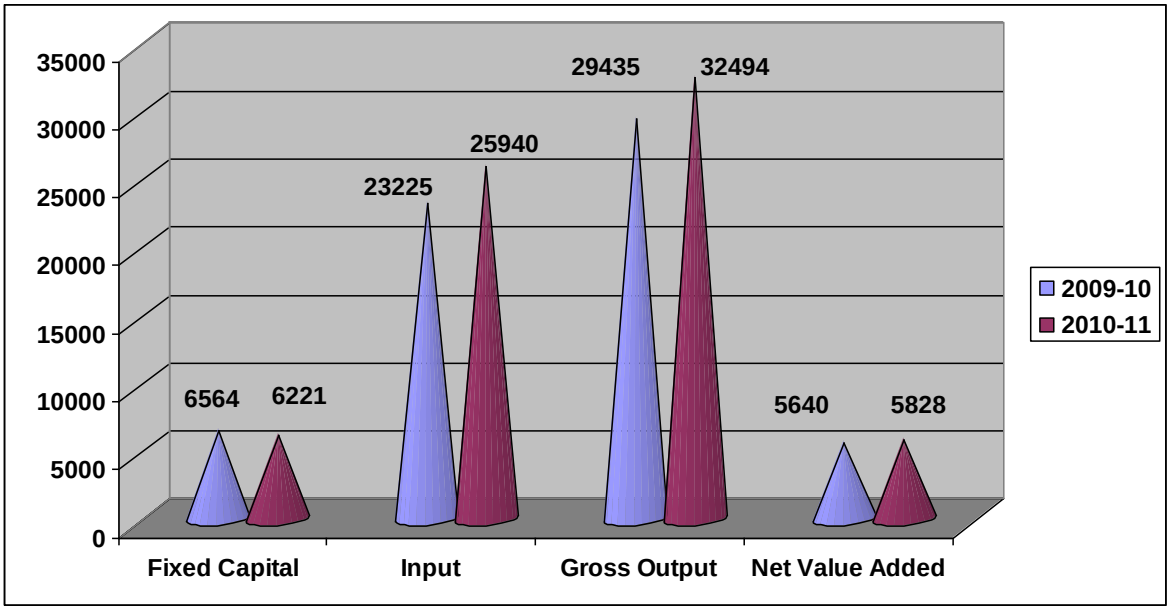
## (ii) No. of workers & Persons engaged




## (iii) Wages and Emoluments (Rs. in crore)



## (iv) Comparison of important characteristics (Rs. in crores)



# CHAPTER – IV

# DISTRIBUTION OF FACTORIES BY SIZE OF EMPLOYMENT

4.1 Table 4.1 reveals the distribution of factories in absolute and percentage terms, by employment range groups as regards certain selected characteristics associated with them. The results for the year 2010-11 are compared with the results of 2009-10.

## TABLE NO. 4.1

### DISTRIBUTION OF FACTORIES BY SIZE OF EMPLOYMENT

**(Rs. in crores.)**

| Employment range | Factories (Number) | | Fixed Capital | | Workers (Number) | | Gross Output | | Net value added | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 |
| **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** | **11** |
| 0-49 | 263 | 327 | 620 | 1034 | 4016 | 3789 | 1594 | 7928 | 247 | 1034 |
| (%) | (51) | (56) | (10) | (17) | (10) | (11) | (5) | (24) | (4) | (18) |
| 50-99 | 107 | 121 | 464 | 899 | 5717 | 5948 | 2631 | 3371 | 672 | 398 |
| (%) | (21) | (21) | (7) | (14) | (15) | (16) | (9) | (10) | (12) | (7) |
| 100-199 | 72 | 59 | 1389 | 625 | 6944 | 5293 | 7445 | 2883 | 852 | 771 |
| (%) | (14) | (10) | (21) | (10) | (18) | (15) | (25) | (9) | (15) | (13) |
| 200-499 | 59 | 53 | 2574 | 1840 | 10591 | 9338 | 8763 | 5933 | 2338 | 975 |
| (%) | (11) | (9) | (39) | (30) | (27) | (26) | (30) | (18) | (42) | (17) |
| 500 & above | 19 | 23 | 1517 | 1823 | 11623 | 11709 | 9002 | 12379 | 1531 | 2649 |
| (%) | (3) | (4) | (23) | (29) | (30) | (32) | (31) | (38) | (27) | (45) |
| **Total** | **520** | **583** | **6564** | **6221** | **38891** | **36076** | **29435** | **32494** | **5640** | **5828** |
| **(%)** | **(100)** | **(100)** | **(100)** | **(100)** | **(100)** | **(100)** | **(100)** | **(100)** | **(100)** | **(100)** |

*Figures in bracket indicate percentages*

Out of the total number of 583 factories covered during 2010-11, factory units employing less than 50 persons were found to be covering 56% or 327 units of the total and this coverage was 51% or 263 units of the total in the year 2009-10. In the units employing persons in the range of 500 & above the utilization of fixed capital has increased from 23% in 2009-10 to 29% in 2010-11 and provided employment to the workers to the tune of 32% during the year 2010-11. In 2010-11 Gross output of 38% was recorded as the highest yield of the factory units that classified under the range 500 & above followed by 24% in the employment range of 0-49. The factories in the range of 500 & above generated highest Net value added during the year in consideration which was worth 45% of the total.

## TABLE NO. 4.2

### PERCENTAGE GROWTH IN FACTORIES BY SIZE OF EMPLOYMENT DURING 2010-11 OVER 2009- 10

| Employment range | Factories | Fixed capital | Workers | Gross output | Net value added |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| 0-49 | 24.3 | 66.8 | -5.7 | 397.4 | 318.6 |
| 50-99 | 13.1 | 93.8 | 4.0 | 28.1 | -40.8 |
| 100-199 | -18.1 | -55.0 | -23.8 | -61.3 | -9.5 |
| 200-499 | -10.2 | -28.5 | -11.8 | -32.3 | -58.3 |
| 500 & above | 21.1 | 20.2 | 0.7 | 37.5 | 73.0 |
| **Total** | 12.1 | -5.2 | -7.2 | 10.4 | 3.3 |

**4.2** Table 4.2 reveals a growth of 12.1% in total number of factories covered during 2010-11 compared to 2009-10. While the number of factories covered in the employment range of 0-49 increased by 24.3%, it decreased by 18.1% in the employment range of 100-199. The maximum growth of 93.8% and 66.8% in Fixed Capital is noticed in the employment range of 50-99 and 0-49 respectively. As regards number of workers, only two employment ranges viz 50-99 & 500 & above have registered a marginal positive growth while the remaining employment ranges have registered a negative growth. The employment range of 0-49 secured a growth of 397.4% in Gross Output followed by 37.5% in 500 & above range over the year 2009-10. NVA shows a significant increase of 318.6% in 0-49 range followed by 73.0% in the 500 & above range.

## TABLE NO. 4.3

### STRUCTURAL RATIOS AND TECHNICAL COEFFICIENTS BY SIZE OF EMPLOYMENT

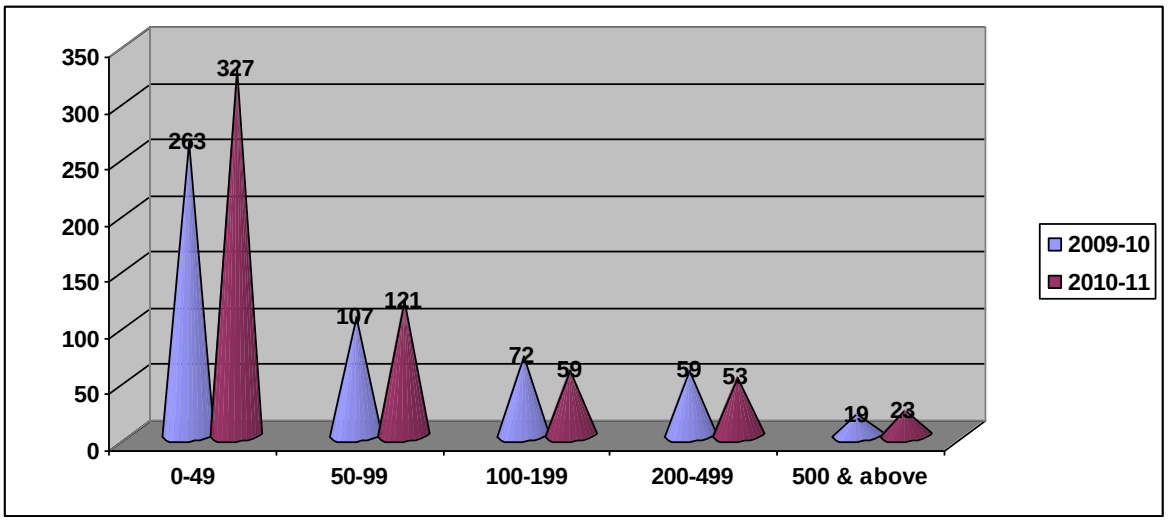
| Employment range | Fixed capital per person engaged (Rs. in lakh) | | Net value added per person engaged (Rs. in lakh) | | Wages per worker (Rs. in lakh) | | Technical Coefficient | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | FC to NVA | | FC to GOP | | NVA to GOP | | Total Emoluments to NVA | |
| | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| 0-49 | 11.63 | 19.87 | 4.64 | 19.87 | 0.86 | 3.11 | 2.51 | 1.00 | 0.39 | 0.13 | 0.16 | 0.13 | 0.28 | 0.28 |
| 50-99 | 6.05 | 10.56 | 8.78 | 4.68 | 0.88 | 1.16 | 0.69 | 2.26 | 0.18 | 0.27 | 0.26 | 0.12 | 0.15 | 0.36 |
| 100-199 | 15.00 | 8.34 | 9.20 | 10.30 | 1.53 | 1.08 | 1.63 | 0.81 | 0.19 | 0.22 | 0.11 | 0.27 | 0.29 | 0.17 |
| 200-499 | 16.96 | 13.35 | 15.41 | 7.07 | 1.27 | 1.39 | 1.10 | 1.89 | 0.29 | 0.31 | 0.27 | 0.16 | 0.16 | 0.30 |
| 500 & above | 10.63 | 10.04 | 10.72 | 14.59 | 1.44 | 1.60 | 0.99 | 0.69 | 0.17 | 0.15 | 0.17 | 0.21 | 0.19 | 0.15 |
| **Total** | **12.70** | **11.70** | **10.91** | **10.96** | **1.27** | **1.56** | **1.16** | **1.07** | **0.22** | **0.19** | **0.19** | **0.18** | **0.19** | **0.22** |

**4.3 Structural ratios and technical coefficient**: Important ratios and coefficients in different employment ranges are given in table 4.3. The average least value of fixed capital utilised during 2010-11 was in the range of 100-199 being Rs.8.34 lakhs, and the highest was in the range of 0-49 being Rs.19.87 lakhs. Highest value of NVA of Rs.19.87 lakhs per person engaged among all the employment ranges was generated in the employment range employing 0-49. The highest average wage remuneration was paid in the employment range of 0-49 i.e. Rs.3.11 lakh during the year 2010-11.

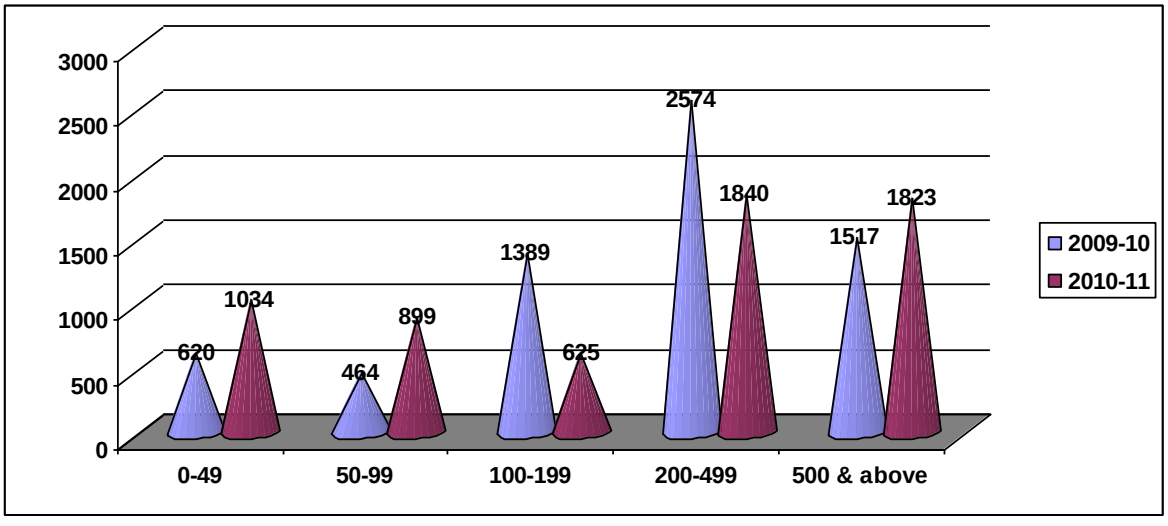
The Net value added co-efficient measured by value of Fixed Capital required to generate each rupee of net value added i.e. (FC to NVA) varies from a low value of 0.69 among factories in employment size class 500 & above to a high value of 2.26 in the employment size class of 50-99 in the year 2010-11, while FC to Gross Output ranged from 0.13 to 0.31 among the factories in the employment range class of 0-49 and 200-499 respectively during the year 2010-11. The value of NVA to Gross output was highest in 100-199 employment range. Total emoluments to NVA were ranged between 0.15 to 0.36 in the employment range of 500 & above and 50-99 respectively. In other words, 15% to 36% of NVA generated was disbursed as emoluments as least and highest remuneration during the current year in the whole set up.

## DISTRIBUTION OF FACTORIES BY SIZE OF EMPLOYMENT

### (i) No. of Factories




### (ii) Fixed Capital (Rs. in crores)




## CHAPTER – V

# DISTRIBUTION OF FACTORIES BY SIZE OF CAPITAL

5.1 **Principal Characteristics**: The distribution of factories together with their share in certain selected characteristics by size of capital, viz. the undepreciated gross value of plant and machinery installed in the factory whether owned or leased in, is given in Table 5.1 along with the percentage distribution of factories according to the investment in Plant and Machinery duly compared with the earlier year.

## TABLE NO. 5.1

### PRINCIPAL CHARACTERISTICS BY SIZE OF CAPITAL

**(Rs. in crore)**

| Capital range of Plant & Machinery (Rs. in lakh) | Factories (Number) | | Fixed Capital | | Workers (number) | | Gross Output | | Net Value Added | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
| Upto 5.0 | 128 | 144 | 229 | 46 | 2572 | 1676 | 211 | 293 | 37 | 50 |
| (%) | (24.62) | (24.70) | (3.49) | (0.74) | (6.61) | (4.65) | (0.72) | (0.90) | (0.66) | (0.86) |
| 5.0 - 20 | 28 | 43 | 12 | 37 | 840 | 719 | 62 | 162 | 19 | 36 |
| (%) | (5.38) | (7.38) | (0.18) | (0.59) | (2.16) | (1.99) | (0.21) | (0.50) | (0.34) | (0.62) |
| 20 - 50 | 60 | 41 | 32 | 58 | 1877 | 783 | 243 | 220 | 36 | 35 |
| (%) | (11.54) | (7.03) | (0.49) | (0.93) | (4.83) | (2.17) | (0.83) | (0.68) | (0.64) | (0.60) |
| 50 - 200 | 92 | 91 | 262 | 261 | 3863 | 2967 | 1811 | 1351 | 266 | 197 |
| (%) | (17.69) | (15.61) | (3.99) | (4.20) | (9.93) | (8.22) | (6.15) | (4.16) | (4.72) | (3.38) |
| 200 - 1000 | 111 | 147 | 702 | 946 | 7776 | 8900 | 3538 | 4767 | 682 | 1239 |
| (%) | (21.35) | (25.21) | (10.69) | (15.21) | (19.99) | (24.67) | (12.02) | (14.67) | (12.09) | (21.26) |
| 1000-5000 | 65 | 72 | 1279 | 1540 | 8712 | 8541 | 9319 | 10563 | 1741 | 2111 |
| (%) | (12.5) | (12.35) | (19.49) | (24.75) | (22.40) | (23.68) | (31.66) | (32.51) | (30.87) | (36.22) |
| >5000 | 36 | 45 | 4048 | 3333 | 13251 | 12491 | 14250 | 15139 | 2859 | 2160 |
| (%) | (6.92) | (7.72) | (61.67) | (53.58) | (34.07) | (34.62) | (48.41) | (46.59) | (50.69) | (37.06) |
| **Total** | **520** | **583** | **6564** | **6221** | **38891** | **36076** | **29434** | **32494** | **5640** | **5828** |
| **(%)** | (**100**) | (**100**) | (**100**) | (**100**) | (**100**) | (**100**) | (**100**) | (**100**) | (**100**) | (**100**) |

*Figures in bracket indicate percentages*.

The concentration of as much as 128 and 144 establishments or 24.62% and 24.70% of the total number of registered factories covered during the years 2009-10 and 2010-11 respectively, invested on an average amount of upto Rs.5 lakh in plant and machinery.

As against these, factories with gross investment in plant & machinery between Rs.200-1000 lakh covered 111 and 147 (21.35% and 25.21%) factories in number and used Rs.702 crore and Rs.946 crore (10.69% & 15.21%) of the fixed capital. Further this group engaged workforce of 7776 and 8900; it produced goods and services worth Rs.3538 crore & Rs.4767 crore and contributed Rs.682 crore & Rs.1239 crore on net value added by manufacture during the years 2009-10 and 2010-11 respectively.

The lowest number of factories covered during 2009-10 was 28 or 5.38% in the capital range of Rs.5.0-20 lakhs, and it was 41 or 7.03% during 2010-11 in the capital range of Rs.20-50 lakhs. The factories in the capital range of Rs.5000 lakh and above utilized fixed capital worth Rs.4048 crore & Rs.3333 crores (61.67% and 53.58% each), which provided employment to 13251 and 12491 (34.07% & 34.62%) persons during 2009-10 & 2010-11 respectively. The contribution of the factories of above Rs.5000 lakh capital range towards gross output was to the tune of Rs.14250 crore and Rs.15139 crore or 48.41% & 46.59% and the net value added by manufacture was Rs.2859 crore or 50.69% and Rs.2160 crore or 37.06% respectively during the years compared.

The lowest fixed capital of Rs.12 crore (0.18%) during 2009-10 and Rs.37 crore (0.59%) during the year 2010-11 were noticed in the capital range of Rs.5.0-20 lakh. The least number of workers employed were 840 and 719 workers in the capital range of Rs.5.0-20 lakh during the years 2009-10 & 2010-11 respectively who produced goods/services of the value of Rs.62 crores and Rs.162 crore contributing to the net income in the form of net value added of the order of Rs.19 crore and Rs.36 crore of the total net value added during the years 2009-10 and 2010-11.

## TABLE NO. 5.2

### PERCENTAGE GROWTH IN PRINCIPAL CHARACTERISTICS BY SIZE OF CAPITAL DURING 2010-11 OVER 2009-10

| Capital range of Plant & Machinery (Rs. in lakh) | Factories | Fixed Capital | Workers | Gross Output | Net Value Added |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| Upto 5.0 | 12.5 | -79.9 | -34.8 | 38.9 | 35.1 |
| 5.0 - 20 | 53.6 | 208.3 | -14.4 | 161.3 | 89.5 |
| 20 - 50 | -31.7 | 81.3 | -58.3 | -9.5 | -2.8 |
| 50 - 200 | -1.1 | -0.4 | -23.2 | -25.4 | -25.9 |
| 200 - 1000 | 32.4 | 34.8 | 14.5 | 34.7 | 81.7 |
| 1000-5000 | 10.8 | 20.4 | -2.0 | 13.3 | 21.3 |
| >5000 | 25.0 | -17.7 | -5.7 | 6.2 | -24.4 |
| **Total** | 12.1 | -5.2 | -7.2 | 10.4 | 3.3 |

**Growth:** Table 5.2 exhibits the growth in factories during the year 2010-11 over the previous year in terms of factories with gross investment in plant and machinery. Factories investing in plant and machinery in the capital range of Rs.5.0-20 lakh have registered highest growth of 53.6% followed by 32.4% in Rs.200-1000 lakh capital range. However, there was negative growth in number of factories during 2010-11 over 2009-10 in the capital ranges of Rs.20-50 lakh & Rs.50-200 lakh.

The maximum growth of 208.3% in Fixed capital was noticed in the employment range of Rs.5.0-20 lakh followed by 81.3% in the range of Rs.20-50 lakhs. The highest growth in the number of workers employed was in the capital range of Rs.200-1000 lakh. Gross output grew by 161.3% in the capital range of Rs.5.0-20 lakhs. Maximum growth in net value added i.e.89.5% was observed in the factories investing in plant and machinery in capital range Rs.5.0-20 lakhs. The growth in net value added varied

from -25.9% to 89.5% over the different capital ranges of investment in Plant and Machinery with average growth for all the units registering at 3.3%.

5.3 **Structural Ratios and Technical Coefficients**: Table 5.3 below reveals structural ratios and technical coefficient of factories, classified by size of capital invested in plant and machinery.

## TABLE NO. 5.3

### STRUCTURAL RATIOS AND TECHNICAL COEFFICIENTS BY SIZE OF CAPITAL

| Capital range of Plant & Machinery (Rs. in Lakh) | Fixed capital per person engaged (Rs. in lakh) | | Net value added per person engaged (Rs. in lakh) | | Wages per worker (Rs. in lakh) | | Technical Coefficient | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | FC to NVA | | FC to GOP | | NVA to GOP | | Total Emoluments to NVA | |
| | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| Upto 5.0 | 7.51 | 2.22 | 1.21 | 2.42 | 0.64 | 0.81 | 6.21 | 0.92 | 1.09 | 0.16 | 0.18 | 0.17 | 0.69 | 0.41 |
| 5.0 - 20 | 1.23 | 3.91 | 1.99 | 3.80 | 0.49 | 0.80 | 0.62 | 1.03 | 0.19 | 0.23 | 0.31 | 0.22 | 0.30 | 0.53 |
| 20 - 50 | 1.41 | 5.18 | 1.56 | 3.13 | 0.51 | 1.03 | 0.90 | 1.65 | 0.13 | 0.27 | 0.15 | 0.16 | 0.55 | 0.46 |
| 50 - 200 | 5.06 | 5.62 | 5.12 | 4.23 | 0.91 | 1.21 | 0.99 | 1.33 | 0.14 | 0.19 | 0.15 | 0.15 | 0.26 | 0.36 |
| 200 - 1000 | 6.09 | 7.63 | 5.92 | 9.99 | 0.98 | 1.14 | 1.03 | 0.76 | 0.20 | 0.20 | 0.19 | 0.26 | 0.29 | 0.21 |
| 1000 - 5000 | 10.78 | 11.20 | 14.67 | 15.35 | 1.13 | 2.11 | 0.73 | 0.73 | 0.14 | 0.15 | 0.19 | 0.20 | 0.13 | 0.21 |
| >5000 | 24.07 | 18.31 | 17.01 | 11.87 | 1.91 | 1.73 | 1.42 | 1.54 | 0.28 | 0.22 | 0.20 | 0.14 | 0.19 | 0.20 |
| **Total** | **12.70** | **11.70** | **10.91** | **10.96** | **1.27** | **1.56** | **1.16** | **1.07** | **0.22** | **0.19** | **0.19** | **0.18** | **0.19** | **0.22** |

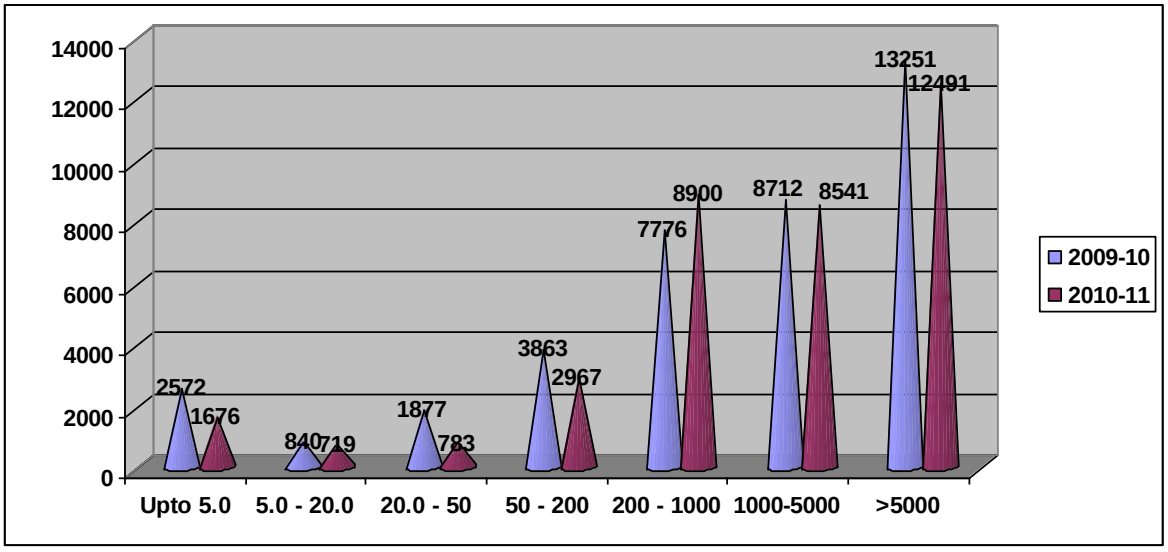
The lowest and the highest ratio for Fixed Capital per person engaged during 2009-10 was Rs.1.23 lakh and Rs.24.07 lakh in the capital range of Rs.5.0-20 lakh and above Rs.5000 lakh respectively, while during 2010-11 it was Rs.2.22 lakh and Rs.18.31 lakh in the capital ranges of upto Rs.5 lakh and above Rs.5000 lakh respectively. The Net value added contributed by each person engaged was as high as Rs.17.01 lakh and Rs.15.35 lakh in the factories of the capital range of above Rs.5000 lakh and Rs.1000- 5000 lakh range for the year 2009-10 and 2010-11 respectively and the lowest being Rs.1.21 lakh and Rs.2.42 lakh in the capital range of Upto Rs.5 lakhs.

The workers employed in the capital range investment of above Rs.5000 lakh & Rs.1000-5000 lakh were paid average highest remuneration to the tune of Rs.1.91 lakh and Rs.2.11 lakh per annum during the respective years, average remuneration of all the capital ranges being Rs.1.27 lakh and Rs.1.56 Lakh respectively for the years under consideration.

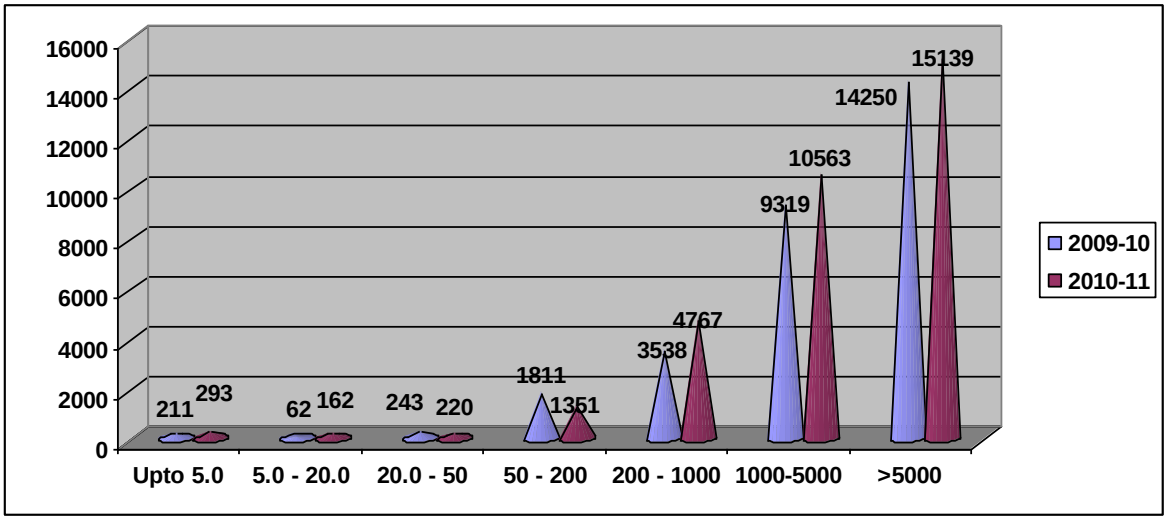
As far as co-efficient is concerned, least Fixed Capital utilized to generate Net Value Added worth a rupee noted during 2009-10 and 2010-11 was 0.62 and 0.73 in the capital range Rs.5.0 - 20 lakh and Rs.1000 - 5000 lakh. However, the highest was in the range of upto Rs.5 lakh i.e. 6.21 during 2009-10 and in the year 2010-11 was 1.65 in the capital range of Rs.20 - 50 lakh. The highest Fixed Capital utilized to produce Gross Output was 0.27 in the range of Rs.20-50 lakh, and the least was 0.15 which falls in the range of Rs.1000-5000 lakh for the year 2010-11. The Highest NVA to GOP was in the range of Rs.200-1000 lakh i.e. 0.26 and lowest was in the range of above Rs.5000 lakh being 0.14 for the current year.

## PRINCIPAL CHARACTERISTICS BY SIZE OF CAPITAL

### (i) No. of workers




### (ii) Gross Output (Rs. in crores)



# CHAPTER – VI

# DISTRIBUTION OF FACTORIES BY TYPE OF OWNERSHIP

**Principal Characteristics** : Table No.6.1 indicate the distribution of factories canvassed during the years 2009-10 and 2010-11 in absolute and percentage terms by type of ownership viz. public, private and joint sectors.

## TABLE NO. 6.1

**Principal Characteristics by Type of Ownership**

**(Value in Rs. crore)**

| Type of Ownership | Factories | | Fixed Capital | | Workers (Number) | | Output | | Net Value Added | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
| 1. Wholly Central Govt * | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| (%) | (0) | (0) | (0) | (0) | (0) | (0) | (0) | (0) | (0) | (0) |
| 2. Wholly State Govt. | 2 | 2 | 5 | 6 | 158 | 146 | 24 | 32 | -36 | 11 |
| (%) | (0.38) | (0.34) | (0.08) | (0.10) | (0.41) | (0.40) | (0.08) | (0.10) | (-0.64) | (0.19) |
| 3. Central & State/local Govt | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| (%) | (0) | (0) | (0) | (0) | (0) | (0) | (0) | (0) | (0) | (0) |
| **4. Public Sector (1+2+3)** | **2** | **2** | **5** | **6** | **158** | **146** | **24** | **32** | **-36** | **11** |
| **(%)** | **(0.38)** | **(0.34)** | **(0.08)** | **(0.10)** | **(0.41)** | **(0.40)** | **(0.08)** | **(0.10)** | **(-0.64)** | **(0.19)** |
| 5. Joint Sector (Public) | 11 | 23 | 495 | 821 | 4086 | 6485 | 1520 | 3190 | 450 | 1225 |
| (%) | (2.12) | (3.95) | (7.54) | (13.20) | (10.51) | (17.98) | (5.16) | (9.82) | (7.98) | (21.02) |
| 6. Joint Sector (Private) | 3 | 15 | 8 | 95 | 235 | 724 | 102 | 347 | 2 | 160 |
| (%) | (0.58) | (2.57) | (0.12) | (1.53) | (0.60) | (2.01) | (0.35) | (1.07) | (0.04) | (2.75) |
| **7. Joint Sector (5+6)** | **14** | **38** | **503** | **916** | **4321** | **7209** | **1622** | **3537** | **452** | **1385** |
| **(%)** | **(2.69)** | **(6.52)** | **(7.66)** | **(14.72)** | **(11.11)** | **(19.98)** | **(5.51)** | **(10.89)** | **(8.01)** | **(23.76)** |
| 8. Wholly Private | 504 | 543 | 6056 | 5299 | 34413 | 28722 | 27789 | 28925 | 5188 | 4432 |
| (%) | (96.92) | (93.14) | (92.26) | (85.18) | (88.48) | (79.62) | (94.41) | (89.02) | (91.99) | (76.05) |
| **Total** | **520** | **583** | **6564** | **6221** | **38891** | **36076** | **29435** | **32494** | **5640** | **5828** |
| **(%)** | **(100.00)** | **(100.00)** | **(100.00)** | **(100.00)** | **(100.00)** | **(100.00)** | **(100.00)** | **(100.00)** | **(100.00)** | **(100.00)** |

*Figures in bracket indicate percentages*

It may be observed that a very large in number viz. 504 and 543 or 96.92% and 93.14% of the total factories belonging to wholly private ownership category, controlled the whole set up dominating in all attributes over all the sectors. This sector alone provided employment to 34413 and 28722 workers during the years under discussion, which works out to 88.48% & 79.62% for both the years, utilized fixed capital worth Rs.6056 crore (92.26%) and Rs.5299 crore (85.18%) and thereby produced goods/services worth Rs.27789 crore (94.41%) and Rs.28925 crore (89.02%) which accounted for increasing the level of income of the state in the form of net value added by Rs.5188 crore (91.99%) and Rs.4432 crore (76.05%) respectively of the total for both the years i.e. 2009-10 and 2010-11.

As against these the public sector factories either owned wholly or jointly by the Central, State or Local Governments were only 2 in numbers (0.38% and 0.34%) of the total factories during the year 2009-10 and 2010-11 respectively. These factories controlled Rs.5 crore and Rs.6 crore or (0.08%) and (0.10%) of the total fixed capital and engaged 158 or (0.41%) and 146 or (0.40%) of the total workforce respectively in the registered factories sector during 2009-10 and 2010-11 respectively. Their shares in the total gross output and net value added were Rs.24 crore and Rs.32 crore or (0.08%) and (0.10%) and Rs.-36 crore or (-0.64%) and Rs.11 crore or (0.19%) of the respective totals.

The third sector i.e. Joint sector covered 14 and 38 factories each, utilized fixed capital worth Rs.503 crore and Rs.916 crore, engaged 4321 and 7209 workers manufactured products of the value of Rs.1622 crore and Rs.3537 crore during both the years. The Net Value Added generated was of the order of Rs.452 crore and Rs.1385 crore respectively during the year 2009-10 and 2010-11.

## TABLE NO. 6.2

### PERCENTAGE GROWTH IN PRINCIPAL CHARACTERISTICS BY TYPE OF OWNERSHIP DURING 2010-11 OVER 2009-10

| Type of Ownership | Factories | Fixed Capital | Workers | Output | Net Value Added |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 |
| 1. Wholly Central Govt. | -- | -- | -- | -- | -- |
| 2. Wholly State Govt. | 0.00 | 20.00 | -7.59 | 33.33 | 1100.00 |
| 3. Central & State/local Govt. | -- | -- | -- | -- | -- |
| 4. Public Sector (1+2+3) | 0.00 | 20.00 | -7.59 | 33.33 | 1100.00 |
| 5. Joint Sector (Public) | 109.09 | 65.86 | 58.71 | 109.87 | 172.22 |
| 6. Joint Sector (Private) | 400.00 | 1087.50 | 208.09 | 240.20 | 7900.00 |
| 7. Joint Sector (5+6) | 171.43 | 82.11 | 66.84 | 118.06 | 206.42 |
| 8. Wholly Private | 7.74 | -12.50 | -16.54 | 4.09 | -14.57 |
| **Total** | 12.12 | -5.23 | -7.24 | 10.39 | 3.33 |

**Growth**: As compared to 2009-10, the number of factories in the wholly private sector during 2010-11 has increased by 39 i.e. (7.74%) but declined in the fixed capital by 12.50%. It also registered a decline of 16.54% in number of workers, produced goods and services with a growth of 4.09%, thereby generating net value added which was lower by 14.57% over 2009-10. Under Public sector, while there was an increase of 20% in the fixed capital and a growth of 33.33% and 1100% in gross output and net value added, the number of workers declined by 7.59%, leaving the growth in the number of factories at constant over the year 2009-10. Similarly, under Joint Sector all categories have shown a positive growth.

# TABLE NO. 6.3

## STRUCTURAL RATIOS AND TECHNICAL COEFFICIENTS BY TYPE OF OWNERSHIP

| Type of Ownership | Fixed capital per person engaged (Rs. in lakh) | | Net value added per person engaged (Rs. in lakh) | | Wages per worker (Rs. in lakh) | | Technical Coefficient | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | FC to NVA | | FC to GOP | | NVA to GOP | | Total Emoluments to NVA | |
| | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| 1. Wholly Central Govt. | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| 2. Wholly State Govt. | 2.12 | 2.50 | -0.14 | 4.86 | 1.22 | 2.02 | -15.14 | 0.52 | 0.23 | 0.17 | -0.01 | 0.34 | -10.28 | 0.50 |
| 3. Central & State/local Govt. | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- | -- |
| **4. Public Sector (1+2+3)** | **2.12** | **2.50** | **-0.14** | **4.86** | **1.22** | **2.02** | **-15.14** | **0.52** | **0.23** | **0.17** | **-0.01** | **0.34** | **-10.28** | **0.50** |
| 5. Joint Sector (Public) | 10.24 | 9.69 | 9.31 | 14.46 | 0.99 | 0.96 | 1.10 | 0.67 | 0.33 | 0.26 | 0.30 | 0.38 | 0.15 | 0.10 |
| 6. Joint Sector (Private) | 2.44 | 7.53 | 0.78 | 12.61 | 1.45 | 1.02 | 3.11 | 0.60 | 0.07 | 0.28 | 0.02 | 0.46 | 2.25 | 0.11 |
| **7. Joint Sector (5+6)** | **9.76** | **9.41** | **8.79** | **14.22** | **1.02** | **0.97** | **1.11** | **0.66** | **0.31** | **0.26** | **0.28** | **0.39** | **0.16** | **0.11** |
| 8. Wholly Private | 13.08 | 12.27 | 11.21 | 10.26 | 1.30 | 1.70 | 1.17 | 1.20 | 0.22 | 0.18 | 0.19 | 0.15 | 0.19 | 0.25 |
| **Total** | **12.70** | **11.70** | **10.91** | **10.96** | **1.27** | **1.56** | **1.16** | **1.07** | **0.22** | **0.19** | **0.19** | **0.18** | **0.19** | **0.22** |

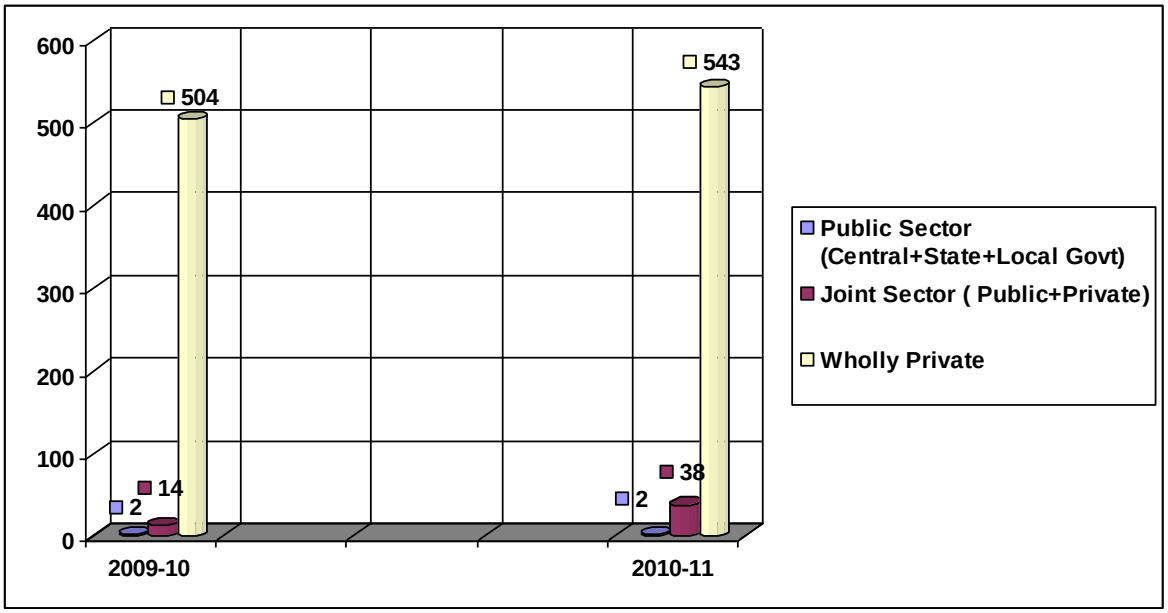
**Structural ratios and technical coefficients**: Certain structural ratios and technical coefficients for each type of ownership are given in table 6.3. On an average each person engaged in wholly private sector generated an income in the form of net value added which was worth Rs.11.21 lakh and Rs.10.26 lakh during both the years. It utilized fixed capital worth Rs.13.08 lakh and Rs.12.27 lakh respectively during 2009-10 & 2010-11, but the remuneration paid to each worker in this profit oriented sector was of the order of Rs.1.30 lakh and Rs.1.70 lakh respectively during both the years.

The net value added for each rupee of gross output produced in public sector has increased from -1 paise in 2009-10 to 34 paise in 2010-11. Each rupee of gross output produced came from utilization of fixed capital worth 23 paise and 17 paise respectively during 2009-10 & 2010-11 in Public Sector and it was 22 paise & 18 paise in wholly private sector during respective years. Similarly, each rupee of net value added has come from utilization of fixed capital worth of Rs.-15.14 and Rs.0.52 respectively during 2009-10 and 2010-11 in public sector, while it was Rs.1.17 and Rs.1.20 in wholly private sector during respective years.

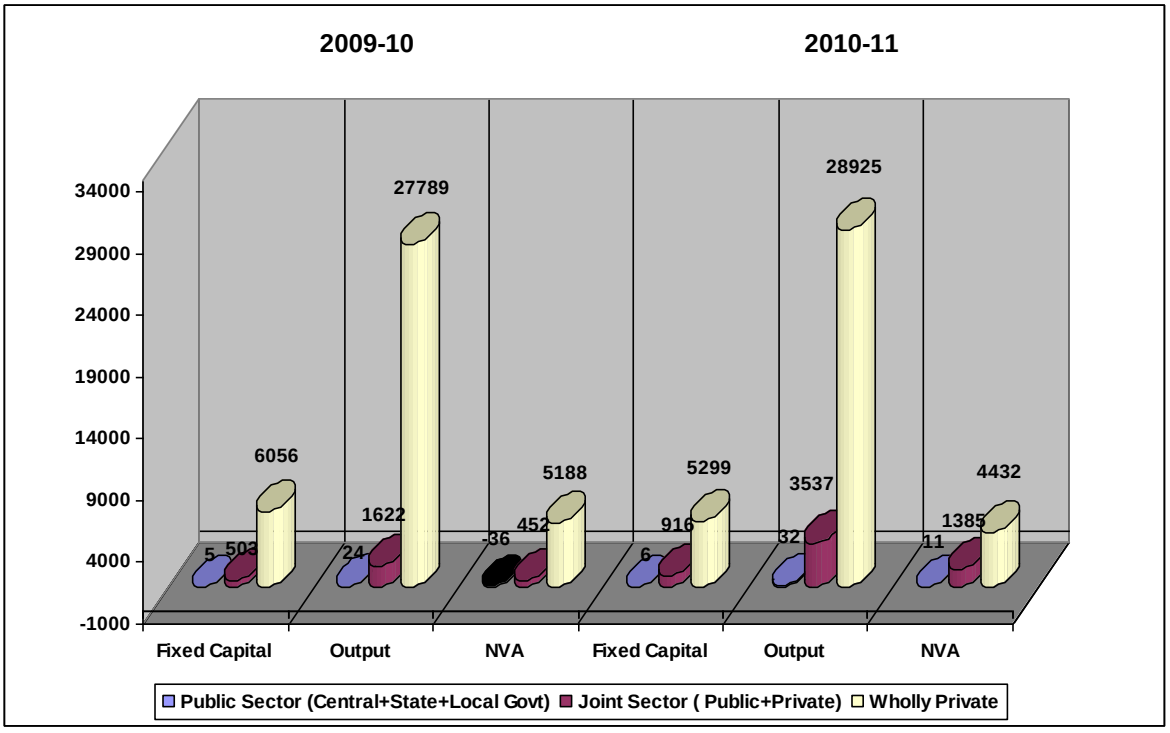
Among all the sectors, the average emoluments paid to a person engaged in production activity in relation to NVA generated was higher in Joint sector private in 2009-10 and in Public sector State during 2010-11.

## Principal Characteristics by Type of Ownership

### (i) No. of Factories




### (ii) Comparison of important characteristics by Type of Ownership



# CHAPTER – VII

# DISTRIBUTION OF FACTORIES BY TYPE OF ORGANISATION

7.1 **Principal Characteristics:** Table 7.1 gives the distribution of factories and other related aggregates in absolute and percentage terms respectively by type of organization. The organizations are classified as (i) Individual proprietorship, (ii) Joint family (iii) Partnership, (iv) Public limited company,(v) Private limited company,(vi) Khadi and village industry (vii) Co-operative Society and ( viii ) others. The public limited company and private limited company constitute the corporate sector.

## Table 7.1

### PRINCIPAL CHARACTERISTICS BY TYPE OF ORGANISATION

**(Value in Rs. Crore)**

| Type of Organisation | | Factories (Number) | | Fixed Capital | | Workers (number) | | Gross Output | | Net Value Added | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| 1 | Individual proprietorship | 79 | 124 | 65 | 41 | 1334 | 1283 | 240 | 247 | 72 | 57 |
| | (%) | (15.19) | (21.27) | (0.99) | (0.66) | (3.43) | (3.56) | (0.82) | (0.76) | (1.28) | (0.98) |
| 2 | Joint Family | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | (%) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) |
| 3 | Partnership | 0 | 68 | 0 | 49 | 0 | 1886 | 0 | 420 | 0 | 77 |
| | (%) | (0.00) | (11.66) | (0.00) | (0.79) | (0.00) | (5.23) | (0.00) | (1.29) | (0.00) | (1.32) |
| 4 | Public Limited Company | 78 | 141 | 54 | 3803 | 2604 | 19643 | 275 | 19677 | 61 | 4057 |
| | (%) | (15) | (24.19) | (0.82) | (61.13) | (6.70) | (54.45) | (0.93) | (60.56) | (1.08) | (69.61) |
| 5 | Private Limited Company | 92 | 248 | 3843 | 2322 | 16691 | 13021 | 14982 | 12099 | 2702 | 1634 |
| | (%) | (17.69) | (42.54) | (58.55) | (37.33) | (42.92) | (36.09) | (50.90) | (37.23) | (47.91) | (28.04) |
| **6** | **Corporate Sector (4+5)** | **170** | **389** | **3897** | **6125** | **19295** | **32664** | **15257** | **31776** | **2763** | **5691** |
| | **(%)** | **(32.69)** | **(66.72)** | **(59.37)** | **(98.46)** | **(49.61)** | **(90.54)** | **(51.83)** | **(97.79)** | **(48.99)** | **(97.65)** |
| 7 | Khadi & Village Industry | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| | (%) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) | (0.00) |
| 8 | Co-operative Society | 264 | 2 | 2588 | 5 | 17726 | 243 | 13811 | 52 | 2800 | 3 |
| | (%) | (50.77) | (0.34) | (39.43) | (0.08) | (45.58) | (0.67) | (46.92) | (0.16) | (49.65) | (0.05) |
| 9 | Others (including Trusts, Wakf Boards, etc.) | 7 | 0 | 14 | 0 | 536 | 0 | 127 | 0 | 5 | 0 |
| | | (1.35) | (0.00) | (0.21) | (0.00) | (1.38) | (0.00) | (0.43) | (0.00) | (0.09) | (0.00) |
| **Total** | | **520** | **583** | **6564** | **6221** | **38891** | **36076** | **29435** | **32494** | **5640** | **5828** |
| **(%)** | | **(100.00)** | **(100.00)** | **(100.00)** | **(100.00)** | **(100.00)** | **(100.00)** | **(100.00)** | **(100.00)** | **(100.00)** | **(100.00)** |

*Figures in brackets indicate percentages*

The corporate sector accounted for 170 and 389 factories or 32.69% and 66.72% of the total number of factories and held fixed capital amounting to Rs.3897 crore and Rs.6125 crore comprising 59.37% and 98.46% of the total fixed capital and employed 19295 and 32664 workers i.e. 49.61% & 90.54% of the total during the years considered. The contribution of this sector to the value of output was of the order of 52% & 98% during the years respectively, while net value added was Rs.2763 crore and Rs.5691 crore (49% & 98%) of the total.

Under the corporate sector, the number of factories classified under Private limited company/ organizations were 17.69% and 42.54% of the total during the year 2009-10 & 2010-11 respectively. They utilized 59% and 37% of the fixed capital, their contribution to the value of output and NVA was 51% and 37% and 48% & 28% respectively and employed 43% & 36% of the total workers to carry out the production activity during the year 2009-10 & 2010-11.

## Table No. 7.2

### PERCENTAGE GROWTH IN PRINCIPAL CHARACTERISTICS BY TYPE OF ORGANISATION DURING 2010-11 OVER 2009-10

| Type of Organisation | Factories | Fixed Capital | Workers | Output | Net Value Added |
|---|---|---|---|---|---|
| **1** | 2 | 3 | **4** | **5** | **6** |
| 1. Individual Proprietorship | 56.96 | -36.92 | -3.82 | 2.92 | -20.83 |
| 2. Joint Family | N/A | N/A | N/A | N/A | N/A |
| 3. Partnership | 0 | 0 | 0 | 0 | 0 |
| 4. Public limited company | 80.77 | 6942.59 | 654.34 | 7055.27 | 6550.82 |
| 5. Private limited company | 169.57 | -39.58 | -21.99 | -19.24 | -39.53 |
| **6. Corporate Sector** | **128.82** | **57.17** | **69.29** | **108.27** | **105.97** |
| 7. Khadi & Village Industry | N/A | N/A | N/A | N/A | N/A |
| 8. Cooperative society | -99.24 | -99.81 | -98.63 | -99.62 | -99.89 |
| 9. Others (including Trusts, Wakf Boards, etc.) | -100.00 | -100.00 | -100.00 | -100.00 | -100.00 |
| **Total** | **12.12** | **-5.23** | **-7.24** | **10.39** | **3.33** |

**Growth:** It may be observed that the corporate sector has dominated the composition of factories and has increased by 129% in number of factories, 57% in fixed capital, 108% in the gross output, 106% in the net value added and 69% in the number of workers has declined during the year 2010-11 over 2009-10. Within the corporate sector, in public limited company sector all the characteristics like no. of factories, fixed capital number of workers, output and net value added has shown positive growth. In Cooperative society sector, there was drastic decline in the characteristics considered during 2010-11 over 2009-10.

# TABLE NO. 7.3

## STRUCTURAL RATIOS AND TECHNICAL COEFFICIENTS BY TYPE OF ORGANISATION

| Type of Organisation | Fixed capital per person engaged (Rs. in lakh) | | Net value added per person engaged (Rs. in lakh) | | Wages per worker (Rs. in lakh) | | Technical Coefficient | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | FC to NVA | | FC to GOP | | NVA to GOP | | Total Emolu-ments to NVA | |
| | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 | 2009-10 | 2010-11 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| 1. Individual Proprietorship | 4.23 | 2.56 | 4.68 | 3.52 | 0.52 | 0.73 | 0.90 | 0.73 | 0.27 | 0.17 | 0.30 | 0.23 | 0.14 | 0.27 |
| 2. Joint Family | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 3. Partnership | 0.00 | 2.14 | 0.00 | 3.33 | 0.00 | 0.80 | 0.00 | 0.64 | 0.00 | 0.12 | 0.00 | 0.18 | 0.00 | 0.32 |
| 4. Public limited company | 1.76 | 13.65 | 2.01 | 14.56 | 0.43 | 1.55 | 0.88 | 0.94 | 0.20 | 0.19 | 0.22 | 0.21 | 0.30 | 0.15 |
| 5. Private limited company | 17.55 | 11.04 | 12.34 | 7.77 | 1.74 | 1.75 | 1.42 | 1.42 | 0.26 | 0.19 | 0.18 | 0.14 | 0.24 | 0.36 |
| 6. Corporate Sector | **15.62** | **12.53** | **11.08** | **11.64** | **1.56** | **1.63** | **1.41** | **1.08** | **0.26** | **0.19** | **0.18** | **0.18** | **0.24** | **0.21** |
| 7. Khadi & Village Industry | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 8. Cooperative society | 10.56 | 1.57 | 11.43 | 0.83 | 1.00 | 1.74 | 0.92 | 1.89 | 0.19 | 0.10 | 0.20 | 0.05 | 0.14 | 2.40 |
| 9. Others (including Trusts, Wakf Boards,etc.) | 2.08 | 0.00 | 0.71 | 0.00 | 1.22 | 0.00 | 2.92 | 0.00 | 0.12 | 0.00 | 0.04 | 0.00 | 2.01 | 0.00 |
| **Total** | **12.70** | **11.70** | **10.91** | **10.96** | **1.27** | **1.56** | **1.16** | **1.07** | **0.22** | **0.19** | **0.19** | **0.18** | **0.19** | **0.22** |

**Structural ratios and technical co- efficient**: A comparison of various ratios and technical coefficients of factories classified by type of organization during 2009-10 and 2010-11 are given in table No. 7.3.

During 2009-10 and 2010-11, in the corporate sector, the fixed capital worth of Rs.15.62 lakh and Rs.12.53 lakh were utilized by each person engaged. Within the corporate sector in the public limited company, the fixed capital worth of Rs.1.76 lakh and Rs.13.65 lakh were utilized by each person engaged and it was Rs.17.55 lakh and Rs.11.04 lakh in the private limited company during 2009-10 and 2010-11 respectively. Wages paid per worker of Rs.1.74 lakh during 2009-10 and Rs.1.75 lakh during 2010-11 was highest for both the years in Private Limited Company. Average emoluments paid in relation to generation of NVA of each rupee has increased from 0.19 in 2009-10 to 0.22 in 2010-11.